

GENERAL DESCRIPTION

READ

The READ command selects the bank from BA0, BA1 inputs and starts a burst read access to an active row. Inputs A0-An (For column addresses, n=A8 for x32, n=A9 for x16, n=A11 for x8), provides the starting column location. When A10 is HIGH, this command functions as an AUTO PRECHARGE command. When the auto precharge is selected, the row being accessed will be precharged at the end of the READ burst. The row will remain open for subsequent accesses when AUTO PRECHARGE is not selected. DQ's read data is subject to the logic level on the DQM inputs two clocks earlier. When a given DQM signal was registered HIGH, the corresponding DQ's will be High-Z two clocks later. DQ's will provide valid data when the DQM signal was registered LOW.

WRITE

A burst write access to an active row is initiated with the WRITE command. BA0, BA1 inputs selects the bank, and the starting column location is provided by inputs A0-An (For column addresses, n=A8 for x32, n=A9 for x16, n=A11 for x8). AUTO-PRECHARGE is determined by A10.

The row being accessed will be precharged at the end of the WRITE burst, if AUTO PRECHARGE is selected. If AUTO PRECHARGE is not selected, the row will remain open for subsequent accesses.

A memory array is written with corresponding input data on DQ's and DQM input logic level appearing at the same time. Data will be written to memory when DQM signal is LOW. When DQM is HIGH, the corresponding data inputs will be ignored, and a WRITE will not be executed to that byte/column location.

PRECHARGE

The PRECHARGE command is used to deactivate the open row in a particular bank or the open row in all banks. BA0, BA1 can be used to select which bank is precharged or they are treated as "Don't Care". A10 determined whether one or all banks are precharged. After executing this command, the next command for the selected bank(s) is executed after passage of the period t_{RP} , which is the period required for bank precharging. Once a bank has been precharged, it is in the idle state and must be activated prior to any READ or WRITE commands being issued to that bank.

AUTO PRECHARGE

The AUTO PRECHARGE function ensures that the precharge is initiated at the earliest valid stage within a burst. This function allows for individual-bank precharge without

requiring an explicit command. A10 to enable the AUTO PRECHARGE function in conjunction with a specific READ or WRITE command. For each individual READ or WRITE command, auto precharge is either enabled or disabled. AUTO PRECHARGE does not apply except in full-page burst mode. Upon completion of the READ or WRITE burst, a precharge of the bank/row that is addressed is automatically performed.

AUTO REFRESH COMMAND

This command executes the AUTO REFRESH operation. The row address and bank to be refreshed are automatically generated during this operation. The stipulated period (t_{RC}) is required for a single refresh operation, and no other commands can be executed during this period. This command is executed at least 8192 times for every T_{REF} period. During an AUTO REFRESH command, address bits are "Don't Care". This command corresponds to CBR Auto-refresh.

BURST TERMINATE

The BURST TERMINATE command forcibly terminates the burst read and write operations by truncating either fixed-length or full-page bursts and the most recently registered READ or WRITE command prior to the BURST TERMINATE.

COMMAND INHIBIT

COMMAND INHIBIT prevents new commands from being executed. Operations in progress are not affected, apart from whether the CLK signal is enabled

NO OPERATION

When $\overline{CS}$ is low, the NOP command prevents unwanted commands from being registered during idle or wait states.

LOAD MODE REGISTER

During the LOAD MODE REGISTER command the mode register is loaded from A0-A12. This command can only be issued when all banks are idle.

ACTIVE COMMAND

When the ACTIVE COMMAND is activated, BA0, BA1 inputs selects a bank to be accessed, and the address inputs on A0-A12 selects the row. Until a PRECHARGE command is issued to the bank, the row remains open for accesses.

COMMAND TRUTH TABLE

Function	CKE		$\overline{CS}$	$\overline{RAS}$	$\overline{CAS}$	$\overline{WE}$	BA1	BA0	A12, A11	
	n - 1	n							A10	A9 - A0
Device deselect (DESL)	H	x	H	x	x	x	x	x	x	x
No operation (NOP)	H	x	L	H	H	H	x	x	x	x
Burst stop (BST)	H	x	L	H	H	L	x	x	x	x
Read	H	x	L	H	L	H	V	V	L	V
Read with auto precharge	H	x	L	H	L	H	V	V	H	V
Write	H	x	L	H	L	L	V	V	L	V
Write with auto precharge	H	x	L	H	L	L	V	V	H	V
Bank activate (ACT)	H	x	L	L	H	H	V	V	V	V
Precharge select bank (PRE)	H	x	L	L	H	L	V	V	L	x
Precharge all banks (PALL)	H	x	L	L	H	L	x	x	H	x
CBR Auto-Refresh (REF)	H	H	L	L	L	H	x	x	x	x
Self-Refresh (SELF)	H	L	L	L	L	H	x	x	x	x
Mode register set (MRS)	H	x	L	L	L	L	L	L	L	V

Note: H=V_{IH}, L=V_{IL} x= V_{IH} or V_{IL}, V = Valid Data.

DQM TRUTH TABLE

Function	CKE		DQMH	DQML
	n-1	n		
Data write / output enable	H	x	L	L
Data mask / output disable	H	x	H	H
Upper byte write enable / output enable	H	x	L	x
Lower byte write enable / output enable	H	x	x	L
Upper byte write inhibit / output disable	H	x	H	x
Lower byte write inhibit / output disable	H	x	x	H

Note:

1. H=V_{IH}, L=V_{IL} x= V_{IH} or V_{IL}, V = Valid Data.
2. x16 options shown. x32 DQM0-DQM3 is similar in function.

FUNCTIONAL DESCRIPTION

The 512Mb SDRAMs are quad-bank DRAMs which operate at 3.3V or 2.5V and include a synchronous interface (all signals are registered on the positive edge of the clock signal, CLK).

Read and write accesses to the SDRAM are burst oriented; accesses start at a selected location and continue for a programmed number of locations in a programmed sequence. Accesses begin with the registration of an ACTIVE command which is then followed by a READ or WRITE command. The address bits registered coincident with the ACTIVE command are used to select the bank and row to be accessed (BA0 and BA1 select the bank, A0-A12 select the row). The address bits A0-An; registered coincident with the READ or WRITE command are used to select the starting column location for the burst access.

Prior to normal operation, the SDRAM must be initialized. The following sections provide detailed information covering device initialization, register definition, command descriptions and device operation.

Initialization

SDRAMs must be powered up and initialized in a predefined manner.

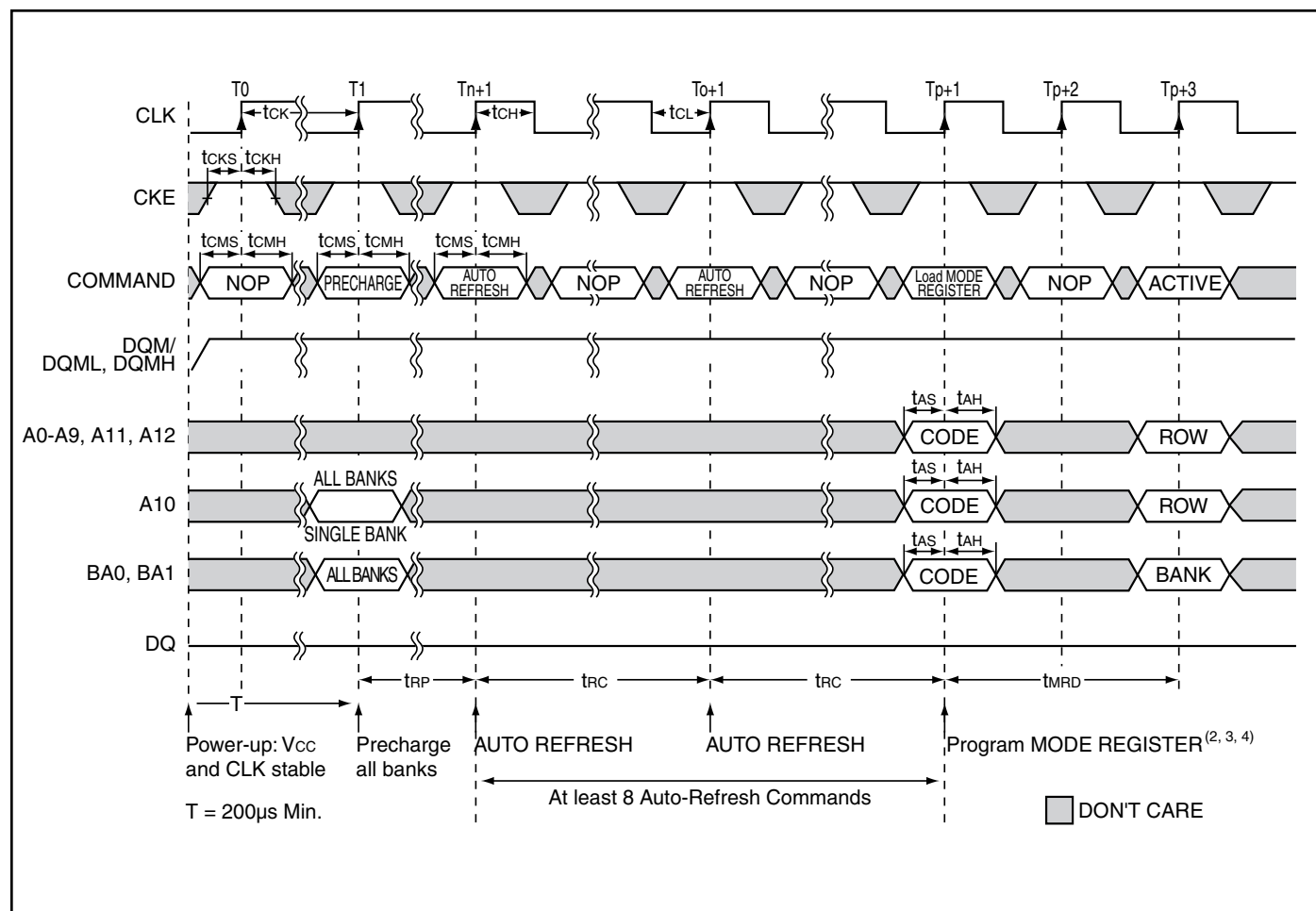
The 512Mb SDRAM is initialized after the power is applied to V_{DD} and V_{DDQ} (simultaneously) and the clock is stable with DQM High and CKE High.

A 100µs delay is required prior to issuing any command other than a COMMAND INHIBIT or a NOP. The COMMAND INHIBIT or NOP may be applied during the 100µs period and should continue at least through the end of the period.

With at least one COMMAND INHIBIT or NOP command having been applied, a PRECHARGE command should be applied once the 100µs delay has been satisfied. All banks must be precharged. This will leave all banks in an idle state after which at least eight AUTO REFRESH cycles must be performed. After the AUTO REFRESH cycles are complete, the SDRAM is then ready for mode register programming.

The mode register should be loaded prior to applying any operational command because it will power up in an unknown state.

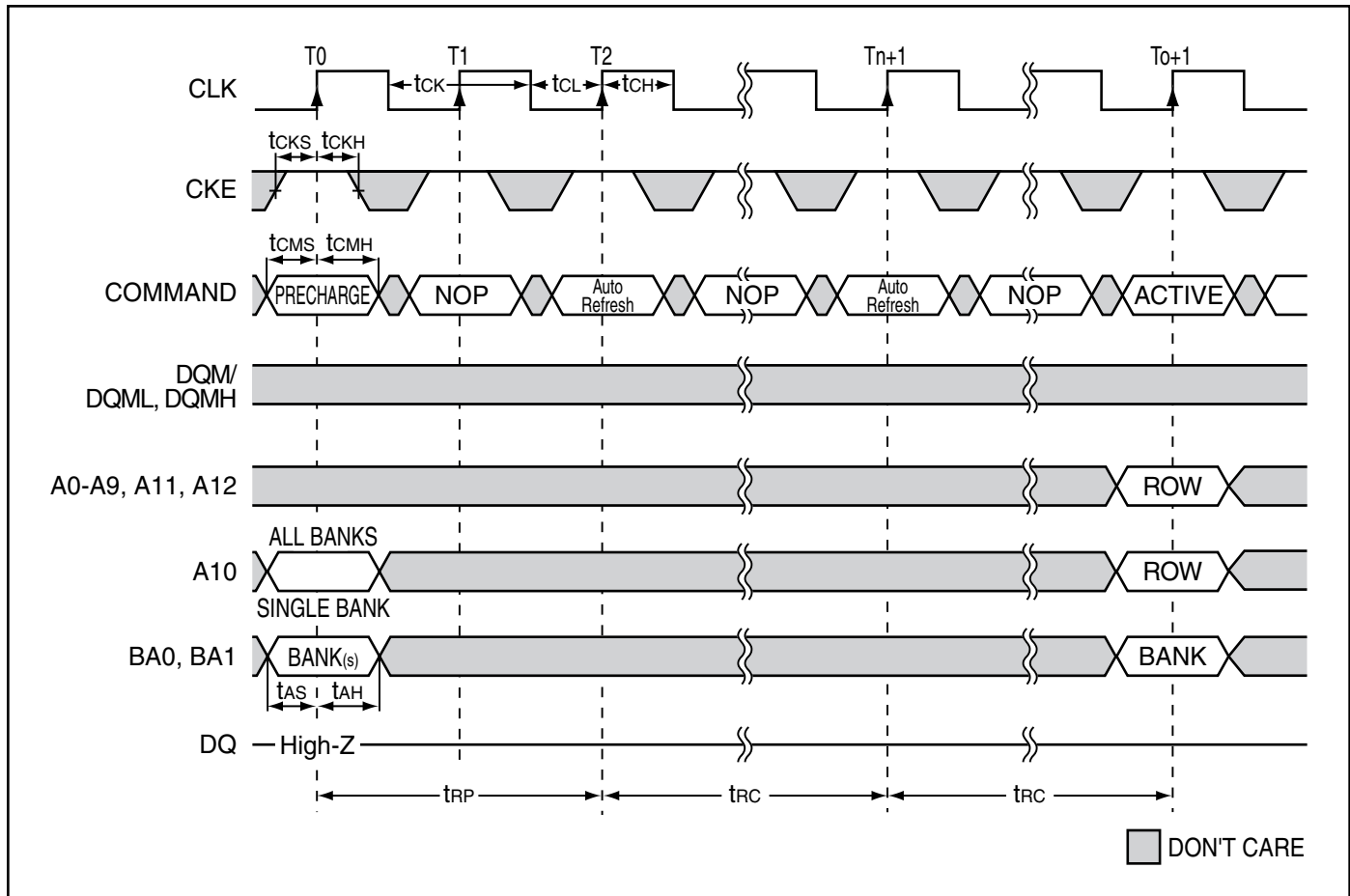
INITIALIZE AND LOAD MODE REGISTER⁽¹⁾



Notes:

1. If $\overline{CS}$ is High at clock High time, all commands applied are NOP.
2. The Mode register may be loaded prior to the Auto-Refresh cycles if desired.
3. JEDEC and PC100 specify three clocks.
4. Outputs are guaranteed High-Z after the command is issued.

AUTO-REFRESH CYCLE



Notes:

1. $\overline{CAS}$ latency = 2, 3